

Elasticity and Its Application

EC107: Microeconomics

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Where we are going today

- ① Why elasticity?
- ② Price elasticity of demand
- ③ Elasticity and total revenue
- ④ Applications
- ⑤ Other elasticities
- ⑥ Robustness and pitfalls

From direction to magnitude

Last lecture we said a leftward supply shift sent the oil price soaring *because demand was inelastic*. That word was doing a lot of work. Today we make it precise.

Elasticity

A measure of how much one variable *responds* to a change in another — the responsiveness of quantity to price, income, or related prices.

Three puzzles that all turn on this single number:

- Will a price cut *raise* or *lower* a firm's revenue?
- Why can a **record harvest** leave farmers *poorer*?
- Why might cracking down on a drug *increase* the crime around it?

By the end, all three have the same one-line answer.

Defining the price elasticity of demand

Definition

$$\varepsilon_d = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

By the law of demand this is negative; by convention we discuss its **absolute value**.

- $|\varepsilon_d| > 1$: **elastic** — quantity responds proportionately *more* than price.
- $|\varepsilon_d| < 1$: **inelastic** — quantity responds proportionately *less*.
- $|\varepsilon_d| = 1$: **unit elastic**.

It is a *ratio of percentages*, hence unit-free: comparable across goods priced in pounds, litres or kilos.

The midpoint method (and why we need it)

Percentages depend on where you start, so the elasticity from £4 to £6 differs from £6 to £4. The fix: divide each change by its *average*.

$$\varepsilon_d = \frac{(Q_2 - Q_1)/[(Q_1 + Q_2)/2]}{(P_2 - P_1)/[(P_1 + P_2)/2]}$$

Worked example

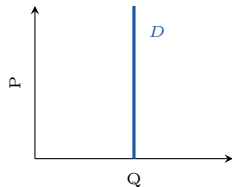
Price £4 → £6; quantity 120 → 80.

$$\% \Delta Q = \frac{80 - 120}{100} = -40\%, \quad \% \Delta P = \frac{6 - 4}{5} = +40\%.$$

$\varepsilon_d = -40\%/40\% = -1 \Rightarrow |\varepsilon_d| = 1$ (unit elastic) — the *same* either direction.

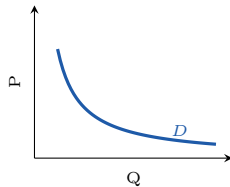
The five degrees of elasticity

Perfectly inelastic ($\varepsilon_d = 0$)



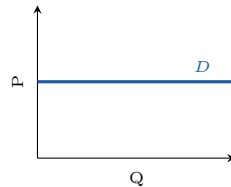
quantity fixed

Unit elastic ($|\varepsilon_d| = 1$)



spending constant

Perfectly elastic ($\varepsilon_d = \infty$)



any rise kills sales

Between the vertical and horizontal extremes lie the merely *inelastic* (steep) and *elastic* (flat) curves.
Flatter = more elastic.

What makes demand elastic?

- **Availability of substitutes** — the biggest factor. Specific brand of butter (elastic) vs. butter in general (less so).
- **Necessity vs. luxury** — insulin (inelastic) vs. cruise holidays (elastic).
- **Market definition** — “food” is inelastic; “vanilla ice cream” is elastic, because narrow categories have close substitutes.
- **Share of budget** — goods that swallow a big slice of income (housing) get more scrutiny than trivial ones (salt, shoelaces).
- **Time horizon** — petrol demand barely moves this week; over years people buy efficient cars, move closer to work, switch to rail.

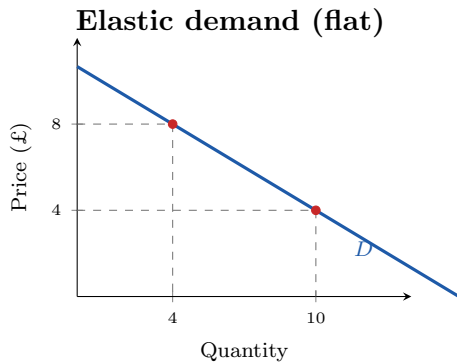
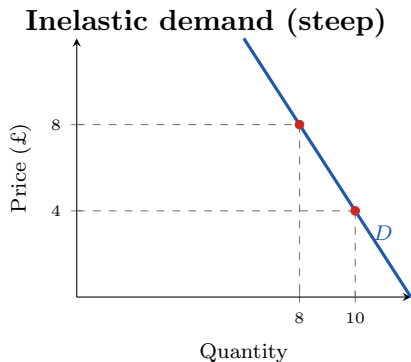
The same good can be inelastic in the short run and elastic in the long run — which is exactly the oil story.

Some illustrative estimates

Good	Approx. $ \varepsilon_d $
Salt, eggs	≈ 0.1 (very inelastic)
Petrol (short run)	≈ 0.25
Petrol (long run)	≈ 0.6
Cars	≈ 1.2
Restaurant meals	≈ 2
Specific brand of cereal	≈ 4 (very elastic)

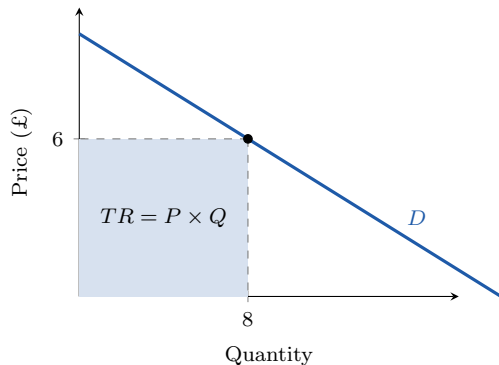
Rough magnitudes only — estimates vary widely by study, market and period. The pattern is the lesson: necessities and broad categories sit low; luxuries, narrow brands and long horizons sit high.

Same price rise, very different responses



Price £4→£8 in both. Inelastic: Q falls 10→8. Elastic: Q falls 10→4.

Total revenue: a tug-of-war



$TR = P \times Q$ — the rectangle under the chosen price. Here $\pounds 6 \times 8 = \pounds 48$.

A price rise sets two effects against each other:

- **Price effect (+):** more revenue per unit.
- **Quantity effect (-):** fewer units.

Formally $\% \Delta TR \approx \% \Delta P + \% \Delta Q = \% \Delta P (1 + \varepsilon_d)$.
The winner is whichever effect is larger — i.e. **elasticity**.

The total-revenue test

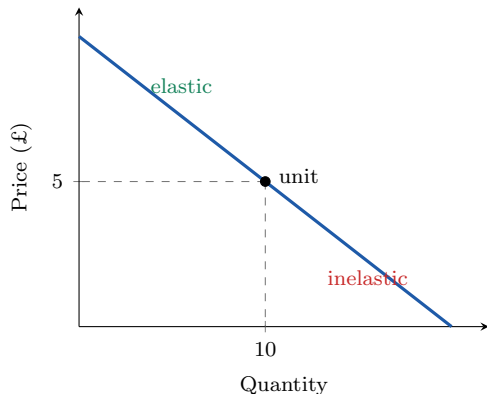
	Price $\uparrow$	Price $\downarrow$
Inelastic ($ \varepsilon_d < 1$)	$TR \uparrow$	$TR \downarrow$
Unit elastic ($ \varepsilon_d = 1$)	TR unchanged	TR unchanged
Elastic ($ \varepsilon_d > 1$)	$TR \downarrow$	$TR \uparrow$

- Inelastic: price effect wins — charge more, earn more.
- Elastic: quantity effect wins — a price *cut* grows revenue.

Manager's rule of thumb

If you don't know your elasticity, you don't know whether your next sale helps or hurts you.

Elasticity changes *along* a straight-line demand curve



Constant *slope*, but elasticity falls as you descend:

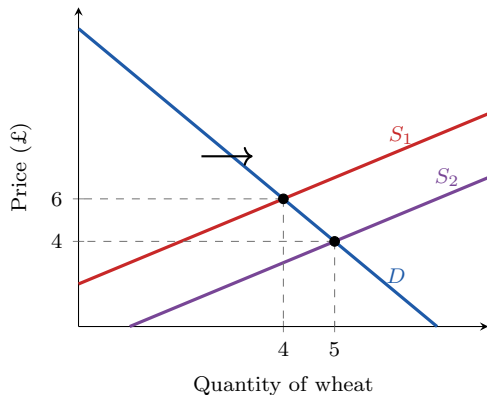
$$|\varepsilon_d| = \frac{1}{|\text{slope}|} \cdot \frac{P}{Q}.$$

High up (P large, Q small) it is elastic; low down, inelastic; **unit elastic at the midpoint** — where **total revenue is maximised**.

Take-away

Slope $\neq$ *elasticity*. A single straight line is elastic, unit, and inelastic in turn.

Puzzle 1 — the bumper-harvest paradox



A great harvest shifts **supply right**. Food demand is *inelastic*, so:

- price falls a lot: £6 → £4;
- quantity rises a little: 4 → 5.

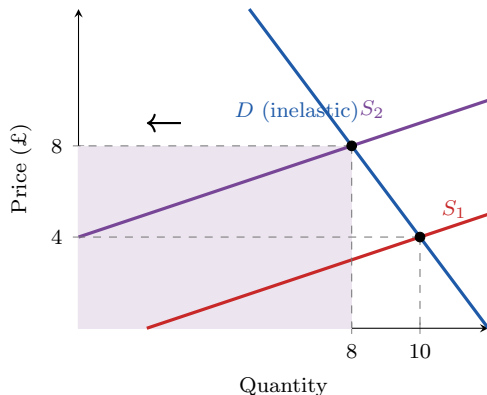
Revenue: £6 × 4 = £24 → £4 × 5 = £20.

Down.

The paradox

Good weather for *all* farmers lowers *every* farmer's income — and explains a century of farm-support policy.

Puzzle 2 — why interdiction can backfire



Interdiction cuts supply ($S_1 \rightarrow S_2$). With *inelastic* demand for an addictive good:

- price jumps £4 $\rightarrow$ £8;
- quantity falls only 10 $\rightarrow$ 8.

Total spending: £40 $\rightarrow$ £64 — it **ris**es (shaded box grows).

The sting

Buyers' total outlay is sellers' total revenue. More money flows to the trade, and addicts may commit *more* crime to fund it. Inelasticity turns a crackdown into a price subsidy for suppliers.

Income elasticity

Income elasticity

$$\varepsilon_I = \frac{\% \Delta Q_d}{\% \Delta \text{income}}$$

- > 0 : **normal** good
- < 0 : **inferior** good
- > 1 : **luxury**; $0 < \varepsilon_I < 1$: **necessity**

Cross-price elasticity

Cross-price elasticity

$$\varepsilon_{AB} = \frac{\% \Delta Q_A}{\% \Delta P_B}$$

- > 0 : **substitutes** (tea, coffee)
- < 0 : **complements** (cars, petrol)
- ≈ 0 : unrelated

Firms use cross-price elasticities to define their real competitors; antitrust authorities use them to define *markets*.

Price elasticity of *supply*

$$\varepsilon_s = \frac{\% \Delta Q_s}{\% \Delta P} \geq 0.$$

Supply is **more elastic** when:

- firms have **spare capacity** and can ramp up cheaply;
- inputs are easy to obtain;
- the good is **storable**;
- the **time horizon** is longer — factories can be built, fields replanted.

Extremes

Perfectly inelastic supply (vertical): a fixed stock — Picasso originals, beachfront land.

Perfectly elastic supply (horizontal): output expandable at constant cost.

Your turn

Your turn

A cinema charges £8 and sells 500 tickets. It considers raising the price to £10, and estimates it would then sell 360.

- ① Compute $|\varepsilon_d|$ over this range (midpoint method).
- ② Is demand elastic or inelastic here?
- ③ Should it raise the price if it cares about revenue? Verify with the actual numbers.

(1) $\% \Delta Q = -140/430 = -32.6\%$, $\% \Delta P = 2/9 = +22.2\%$, $|\varepsilon_d| \approx 1.47$. (2) *Elastic*. (3) *No: revenue £4000 $\rightarrow$ £3600, a fall — consistent with elastic demand.*

Misconceptions & exam traps

- “Steep means inelastic.” Only loosely — elasticity also depends on P/Q , so it *changes along* a straight line. Compare curves only at the same point.
- Treating elasticity as one number for a good. It is **local**: different at every price, and different in the short vs long run.
- Dropping the sign on cross-price/income elasticities. There the *sign* carries the economics (substitute vs complement; normal vs inferior).
- Forgetting the midpoint method and reporting different answers up vs down.
- Confusing total revenue with profit. The TR test is about revenue; costs are a separate story.

Key takeaways

- Elasticity measures **responsiveness**; compute it with the *midpoint method*.
- Demand is more elastic with close substitutes, for luxuries, in narrow markets, for big budget shares, and over longer horizons.
- **TR test**: inelastic $\Rightarrow$ price and revenue move together; elastic $\Rightarrow$ they move apart.
- Slope $\neq$ elasticity — it varies *along* a linear demand curve.
- Inelasticity explains both the bumper-harvest and the interdiction paradoxes.

Coming next

Government policies: price controls and taxes. Elasticity will decide **who really pays** a tax — and how much surplus is destroyed.

Questions?

EC107: Microeconomics — Day 4